MASTER PLAN · v1 · 2026

Polymarket BTC 5-Minute Bot

A phased, design-first build for the up/down prediction markets on Polygon, settled by Chainlink BTC/USD.

OWNER	STARTED	VENUE	HOSTING
Hamad	2026-05-04	Polymarket CLOB	AWS Dublin

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How to read this plan

Phase gates
Concrete success criteria. Don't advance until they're met.

Risk severity
HIGH blocks live trading until mitigated. MED needs a runbook. LOW is tracked.

Decision points
Bold-marked items in **Next 60 Minutes** need your sign-off before Phase 0 starts.

Living document
Mirror this into the Obsidian vault under /runbooks/MASTERPLAN.md and update at every gate.

Executive Summary

The thesis in 60 seconds

Polymarket runs a continuous series of 5-minute binary markets on whether BTC closes higher or lower than its starting price. Markets are deterministic with windows aligned to **floor(now / 300) * 300** and slugs **btc-updown-5m-{window_ts}**. Resolution is the Chainlink BTC/USD snapshot at **window_ts + 300** on Polygon.

The exploitable inefficiency: Chainlink BTC/USD on Polygon uses a **60-second heartbeat with no deviation threshold**. Spot can move 0.3-0.5% while the resolution oracle is silent. Pair that with thin orderbooks ($5K-$50K per window) and the empirical fact that ~15-20% of windows resolve based on the final 10 seconds, and there is a real edge if your infrastructure can read all three signals (spot, oracle, book) faster than the median trader.

60s
Chainlink heartbeat (no deviation trigger)

1-5ms
Dublin VPS round-trip to CLOB

$4.75
Polymarket minimum order size

55-60%
Backtest target win rate after fees

Where the edge comes from, ranked



Where the edge is NOT

- TA on a 5-minute timeframe — RSI/EMA on 1-min candles is mostly noise.
- Manual trading — humans cannot react fast enough at T-10s.
- Deep-volume markets — the inefficiency lives in thin orderbooks.
- Raw latency racing in eu-west-2 — co-located rivals beat you on round-trips; your edge is signal quality and last-second timing.

Tooling Layout

The decided stack

Core stack

Layer	Choice	Why
Language	Python 3.11+	py-clob-client + NautilusTrader native; speed-to-correctness wins for 5-min cadence.
Trading engine	NautilusTrader	Production-grade, deterministic event-driven, official Polymarket adapter, real backtest engine.
Order client	Polymarket/py-clob-client	Official; handles L1 EIP-712 + L2 HMAC-SHA256 signing.
Spot data	Binance + Coinbase WS	Free, low-latency, redundant feeds for BTC/USDT and BTC/USD.
Oracle truth	Chainlink BTC/USD on Polygon	Read via Chainstack / Alchemy / QuickNode RPC.
Storage	SQLite + Parquet	trades.db for fills/PnL, ticks/*.parquet for archive. No external deps.
VPS	AWS Lightsail eu-west-1 Dublin	Sub-5ms to CLOB (London eu-west-2). Avoids UK geo-block.
Memory vault	Obsidian + obsidian-claude-code-mcp	All research / runbooks / postmortems live in vault, MCP-readable.
Risk	Kelly fraction + per-window cap + daily breaker	Standard, audited.
Monitoring	Prometheus + Grafana Cloud + Discord	Free, ops-grade, mobile-friendly alerts.
Secrets	direnv + age-encrypted backup	No cloud secret manager required at this scale.
Process supervisor	systemd unit	Auto-restart, journalctl logs, no extra install.

Claude Code integrations

Tool	Use
Claude Code (this CLI)	Primary dev environment.
obsidian-claude-code-mcp	Read/write the memory vault. Every research note, runbook, postmortem.
Polymarket/agent-skills	Drop into ~/.claude/skills/ — official skills for auth, orders, WS, token ops, bridging, neg-risk, gasless.
gh CLI	Pulling reference repos: py-clob-client, NautilusTrader, etc.
/init	Generate the project's CLAUDE.md once layout is on disk.
/security-review	Run before each phase that touches signing or fund movement.
No autonomous trading via MCP	All live trading runs from the bot binary on Dublin VPS, never from Claude Code.

Polymarket-specific MCPs (research / paper only)

Server	Use
caiovicentino/polymarket-mcp-server	45 tools, real-time WS, exposure / order-size limits.
artvandelay/polymarket-agents	Research MCP (10 tools): odds, orderbook, spread, history.
berlinbra/polymarket-mcp	Read-only market data MCP (lighter).
agent-next/polymarket-paper-trader	Paper-trading simulator with order books and backtests.

Architecture

One diagram, three loops

Three concurrent loops keep the bot honest: a **resolution loop** (per-second) that answers "what does this market resolve as if it ended now," an **execution loop** (T-30s, T-10s, T-5s deadlines) that fires orders by deadline and never skips, and a **supervisor loop** (every minute) that runs heartbeat checks, balance reconciliation, and kill-switch evaluation.



Your laptop · Claude Code session

Obsidian vault <- obsidian-claude-code-mcp <- Claude Code

/runbooks /research /postmortems /strategy-notes /daily

Read/write only. NO live trading from laptop.

Why this shape: the laptop never trades — it edits the vault and reviews paper PnL. The Dublin VPS owns the live execution path. Storage and observability live alongside the bot so a network blip can't lose data or alerts.

Phased Plan

Each phase has a gate. No skipping.

0 — Foundations ~½ day
Gate: Claude can read/write Obsidian; references cloned; CLAUDE.md committed.

1 — Wallet & Auth ~½ day
Gate: Sign requests, read account state, place + cancel a $1 mainnet order.

2 — Read-only Data Plane 1-2 days
Gate: 24h clean capture for spot, oracle, market WS. No gaps.

3 — Backtest Harness 2-3 days
Gate: ≥55% win-rate after fees on ≥500 simulated trades.

4 — Paper Trading Live 1 week
Gate: Paper PnL within 20% of backtest. Zero unhandled exceptions / 7 days.

5 — VPS Provisioning ~½ day
Gate: systemctl healthy; latency p99 < 15ms confirmed.

6 — Live with Kill-switches 1 week
Gate: 7 days live, all kill-switches tested, zero unexplained losses.

7 — Scale + Maker Rebates ongoing
Gate: Stepped scaling: $100 → $500 → $2k → $10k after each positive 7d.

8 — Hardening ongoing
Gate: USDC migration handled, warm standby in Frankfurt, multi-account.

PHASE 0 — Foundations (laptop)

- Install Python 3.11, uv (or poetry), Node (for Obsidian MCP).
- Install Obsidian; create vault polymarket-bot/. Folders: runbooks/, research/, postmortems/, strategy-notes/, daily/.
- Wire iansinnott/obsidian-claude-code-mcp to Claude Code. Verify read/write.
- Clone reference repos into references/: py-clob-client, agent-skills, agents, poly-market-maker, Archetapp gist, evan-kolberg backtesting, aulekator 15m bot.
- Drop Polymarket/agent-skills into ~/.claude/skills/.
- Run /init to seed CLAUDE.md.

PHASE 1 — Wallet & Auth (laptop)

- Create a fresh dedicated Polygon EOA. Never reuse a personal wallet.
- Fund with ~$5 MATIC for gas + small USDC test deposit ($20).
- Use setup_creds pattern from py-clob-client to derive L2 creds (apiKey, secret, passphrase).
- Store secrets in .envrc (gitignored) and back up an age-encrypted copy.
- Read your own positions; place + cancel a $1 limit order on a non-time-critical market.

PHASE 2 Read-only Data Plane (laptop)

- market_feed.py: subscribe to active 5m BTC market via WS, archive to Parquet.
- spot_feed.py: Binance + Coinbase WS for BTC, archive ticks.
- oracle_watcher.py: subscribe to Chainlink BTC/USD AnswerUpdated events on Polygon.
- Sanity check: at any time, answer "resolves as if now" within 50ms of last spot tick.

PHASE 3 Backtest Harness (laptop)

- Wire NautilusTrader's Polymarket adapter (or fork evan-kolberg) to captured data.
- v0 strategy: window-delta + oracle-vs-spot gap. Trade only when |signal| > threshold AND time-to-close in [10s, 30s].
- Realistic fills using top-5 orderbook history, $4.75 minimum, slippage model.
- Run on 30 days. Report win-rate, edge, Sharpe, max drawdown, P50/P95 fill latency.

PHASE 4 Paper Trading Live (laptop, 1 week min)

- Add mode: paper switch — live data, shadow SQLite book, fill marks.
- Run 7 calendar days unattended.
- Diagnose any divergence > 20% between paper PnL and backtest expectation.

PHASE 5 VPS Provisioning

- AWS Lightsail eu-west-1 Dublin, 2 vCPU / 4GB / 80GB NVMe, Ubuntu 24.04.
- Harden: ssh-key only, ufw, fail2ban, auto security updates, non-root user.
- Install Python 3.11, uv, redis, prometheus-node-exporter.
- Latency check: mtr clob.polymarket.com; wscat round-trip. p50 < 5ms, p99 < 15ms.
- systemd unit with restart policy; journalctl 30d retention.
- Prometheus → Grafana Cloud, Discord webhook for unhealthy state.

PHASE 6 Live with Kill-switches

- Initial bankroll $100. Per-trade 1% (~$1). Daily loss circuit breaker -$10.
- Three kill-switches: (a) any feed silent > 10s halts entries. (b) avg fill slippage > 2¢ halts. (c) -$10 daily or -$25 total = systemd stop + Discord page.
- Daily ritual: live PnL vs paper-mode prediction. Investigate every >$3 divergence.

PHASE 7 Scale + Maker Rebates (ongoing)

- Scale only after a positive 7-day live run. Steps: $100 → $500 → $2k → $10k. No skipping.
- Maker layer: park resting limits near mid on weak-signal windows; earn the daily rebate.
- Add a second strategy variant in shadow mode; winning variant takes capital (compete-and-allocate pattern).
- Quarterly postmortem in Obsidian; refit weights on last 30 days.

PHASE 8 Hardening (ongoing)

- Migrate USDC.e → native Polymarket USD as the rollout completes.
- Second VPS in Frankfurt as warm standby; DB replicates via litestream.
- Multi-account proxy wallets for size > $50k (per-account exposure caps).
- On-call rotation with phone alerts.

Risk Register

Read every phase. Update each postmortem.

#	Severity	Risk	Mitigation
01	HIGH	**Wallet compromise** A leaked private key drains everything instantly.	Fresh wallet only. age-encrypt the key. Never paste in chat. Never check in. Hardware wallet for owner key once size > $5k.
02	HIGH	**Geo-restriction** Polymarket geofences UK + US + others.	Run from Dublin (eu-west-1), never eu-west-2 or US regions. Don't connect personal wallet via UI from blocked geos.
03	HIGH	**Adverse selection** Co-located bots in eu-west-2 beat raw round-trips.	Compete on signal quality + last-10-second timing, not raw latency. Track fill-vs-mid slippage daily.
04	MED	**Settlement migration** USDC.e → native USDC ("Polymarket USD") rollout in progress.	Bot must double-write/read during transition. Subscribe to Polymarket announcements; keep migration runbook in Obsidian.
05	MED	**Oracle reorg / stuck oracle** Polygon reorgs are rare but possible; Chainlink can miss a heartbeat.	Design assumes "last answer at end-time" semantics. If oracle stale > 90s, halt new entries.
06	MED	**Min-order math** 5-share × $0.95 = $4.75 minimum spend per limit order.	Don't try near-the-money limits with $100 bankroll and 1% sizing. Use takers until size > $20.
07	LOW	**Tax / regs** Per-trade tax tracking is jurisdiction-dependent.	Export trades.db to CSV monthly. Categorize as appropriate; consult an accountant once PnL > 4-figure.

Next 60 Minutes

Decide and ship Phase 0

- **Confirm the plan above is what you want.** Push back on any choice before code is written.
- **Paste the three tweet texts** (X.com server-side fetch is paywalled). They'll be folded into the right phase.
- **Decide bankroll cap for Phase 6** ($100 default, can go lower to $50).
- **Decide Obsidian vault location** (default: ~/Documents/polymarket-bot-vault/).

After sign-off: Phase 0 begins — clone references, set up Obsidian MCP, scaffold the project. Pause at every phase gate for review.

Sources & References

Canonical links used to compile this plan

Polymarket — Official

Polymarket Developer Docs	docs.polymarket.com/developers/
py-clob-client (Python, primary)	github.com/Polymarket/py-clob-client
rs-clob-client (Rust)	github.com/Polymarket/rs-clob-client
agent-skills (drop into Claude)	github.com/Polymarket/agent-skills
agents (TS autonomous framework)	github.com/Polymarket/agents
poly-market-maker (reference MM)	github.com/Polymarket/poly-market-maker
Authentication (L1 + L2)	docs.polymarket.com/developers/CLOB/authentication
Maker Rebates Program	help.polymarket.com/en/articles/13364471

Reference bot implementations

Archetapp 5m bot gist (best timing/window logic)	gist.github.com/Archetapp/7680adabc48f812a561ca79d73cbac69
aulekator 15m production-grade bot	github.com/aulekator/Polymarket-BTC-15-Minute-Trading-Bot
ThinkEnigmatic adaptive 5m arena	github.com/ThinkEnigmatic/polymarket-bot-arena
evan-kolberg NautilusTrader backtest adapters	github.com/evan-kolberg/prediction-market-backtesting

Infrastructure & oracle

NautilusTrader Polymarket integration	nautilustrader.io/docs/latest/integrations/polymarket/
QuantVPS — Polymarket server location	quantvps.com/blog/polymarket-servers-location
Chainlink BTC/USD on Polygon (60s heartbeat)	data.chain.link/feeds/polygon/mainnet/btc-usd
USDC.e to native USDC migration (Block)	theblock.co/post/388670

Claude tooling

iansinnott/obsidian-claude-code-mcp	github.com/iansinnott/obsidian-claude-code-mcp
MCPVault (Obsidian MCP, no plugin)	medium.com/@ai_transfer_lab/mcpvault-...
MarkusPfundstein/mcp-obsidian	github.com/MarkusPfundstein/mcp-obsidian
caiovicentino/polymarket-mcp-server	github.com/caiovicentino/polymarket-mcp-server
artvandelay/polymarket-agents	github.com/artvandelay/polymarket-agents
berlinbra/polymarket-mcp	github.com/berlinbra/polymarket-mcp
agent-next/polymarket-paper-trader	github.com/agent-next/polymarket-paper-trader

Strategy / edge research

Benjamin-Cup — 5-min market edges (Medium)	medium.com/@benjamin.bigdev/...db8efcb5c196
ILLUMINATION — Beyond simple arbitrage (Medium)	medium.com/illumination/...ddacc92c5b4f
FinanceMagnates — Prediction markets are bot playground	financemagnates.com/trending/...
CoinMarketCap — Polymarket 5-Min BTC launch	coinmarketcap.com/academy/article/polymarket-debuts-5-minute-...

This document is a living plan. Mirror it into the Obsidian vault under /runbooks/MASTERPLAN.md and update at every phase gate.